

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 06



By - Abhishek Sir



Recap of Previous Lecture



Topic

Sliding Window Protocol





Topics to be Covered



Topic

Sliding Window Protocol

Topic

Go Back N ARQ

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50 microsecond. Acknowledgement packets (sent only from B to A) are very small and require negligible transmission time. The propagation delay over the link is 200 microsecond. What is the maximum achievable throughput in this communication?

[GATE 2003]

- (A) $7.69 * 10^6$ bytes per sec
- ✓ (B) $11.11 * 10^6$ bytes per sec
- (C) $12.33 * 10^6$ bytes per sec
- (D) $15.00 * 10^6$ bytes per sec

Ans: B

Solution:-

$$t_x = 50 \mu s$$

$$t_p = 200 \mu s$$

$$\begin{aligned} \text{Cycle time} &= (t_x + 2 * t_p) = 450 \mu s \\ &= (50 + 2 * 200) \mu s \end{aligned}$$

$$\text{Window Size (N)} = 5$$

$$\text{Packet Size} = 1000 \text{ bytes}$$

For Sliding Window ARQ :

$$\text{Throughput} = \frac{\text{Window Size} * \text{Packet Size}}{\text{Cycle Time}} = \frac{5 * 1000 \text{ bytes}}{450 \mu\text{s}}$$

$$= \frac{100}{9} * 10^6 \text{ bytes/sec} = 11.11 * 10^6 \text{ bytes/sec}$$

$$\eta = \frac{N * t_x}{\text{cycle time}} = \frac{5 * 50 \mu\text{s}}{450 \mu\text{s}} = \frac{5}{9}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}}$$

$$\text{Bandwidth} = \frac{1000 \text{ bytes}}{50 \mu\text{s}}$$

$$\text{Throughput} = \eta * \text{Bandwidth}$$

$$= \frac{5}{9} * \frac{100}{5} \text{ bytes}/\mu\text{s}$$

$$= 11.11 * 10^6 \text{ bytes/sec}$$



Topic : Sliding Window ARQ



→ Transmitter's transmitting window size = N $[N > 1]$

→ Receiver's receiving window size = N

→ Total number of sequences = N $[0 \text{ to } (N-1)]$

Total number of sequences
= Transmitter's transmitting window size

Sequence number $\leftarrow$ (Frame number) mod N



Topic : Sliding Window ARQ



Suppose $N = 4$

Sequence Number = 0 to 3

F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}
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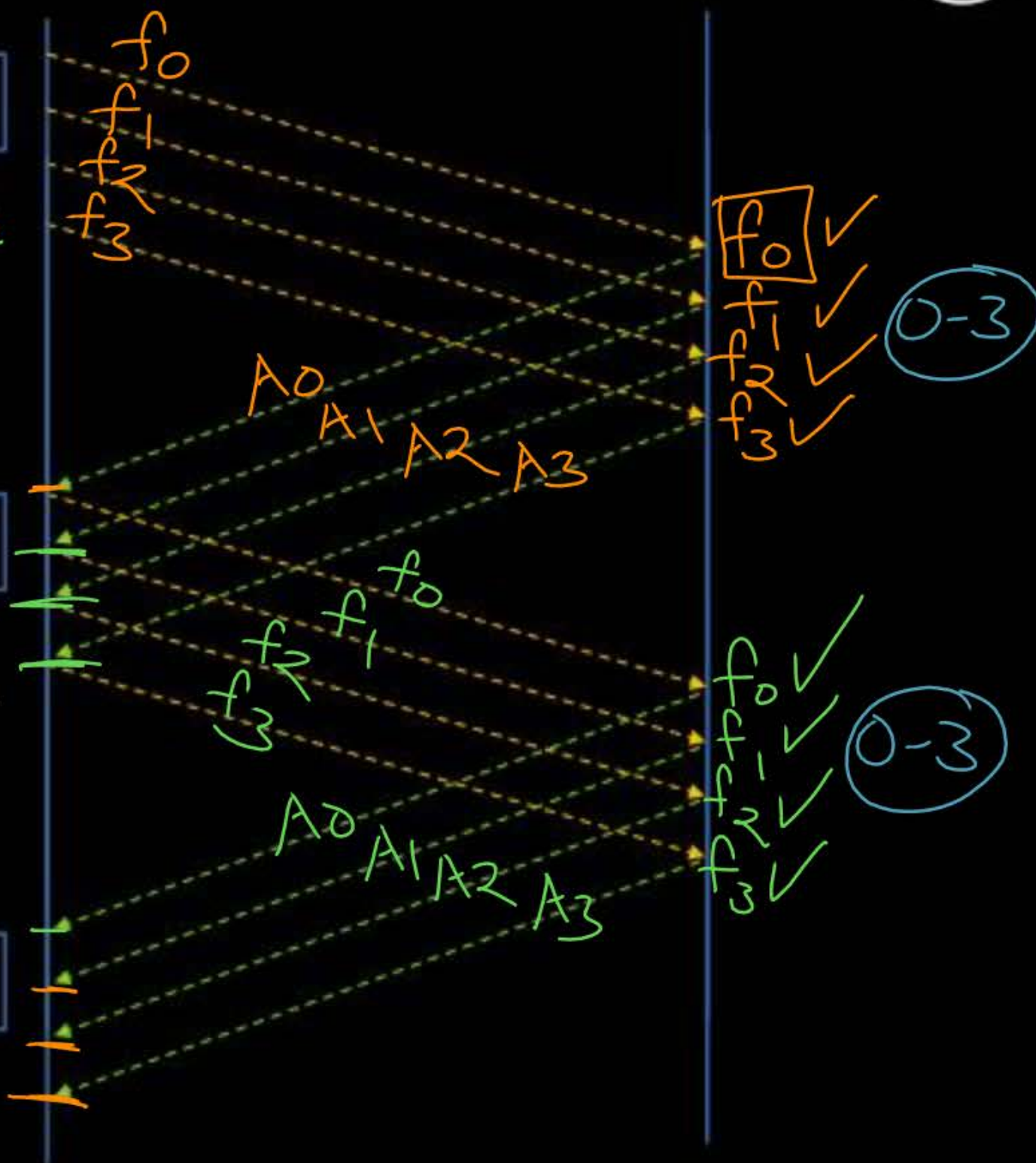
F_0	F_1	F_2	F_3
f_0	f_1	f_2	f_3

F_4	F_5	F_6	F_7
f_0	f_1	f_2	f_3

F_8	F_9	F_{10}	F_{11}
f_0	f_1	f_2	f_3

Transmitter

Receiver





Topic : Sliding Window ARQ

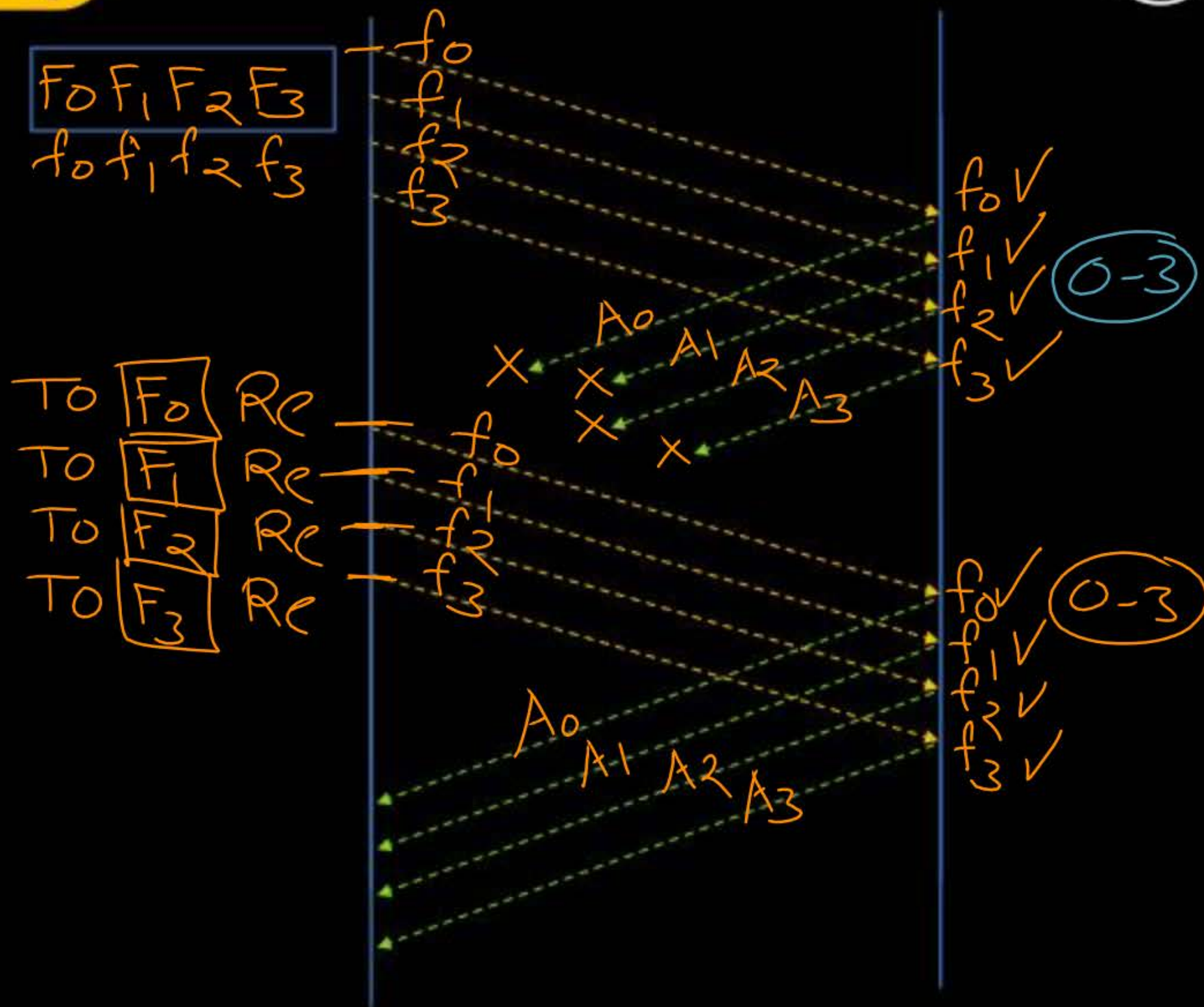


Suppose $N = 4$

Sequence Number = 0 to 3

Prob. 1:-

* Receiver is not able to recognize Duplicate frame set





Topic : Sequence Number



Minimum number of bits required for sequence number field

$$= \lceil \log_2 [\text{Total number of sequences}] \rceil \text{ bits}$$

#Q. The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay (per kilometer) is t seconds. Let R bits/second be the channel capacity. Assuming that processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window is used, is

[GATE 2007]

Solution:-

$$\text{Frame Size} = K \text{ bits}$$

$$\text{Bandwidth} = R \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{K \text{ bits}}{R \text{ bits/sec}} = \frac{K}{R} \text{ sec}$$

$$\text{Distance} = L \text{ Km}$$

$$\text{Signal Speed} = t \text{ Sec/Km}$$

$$t_p = \text{Distance} * \text{Signal Speed} = L \text{ km} * t \text{ sec/km} = Lt \text{ sec}$$

$$\text{Cycle time} = (t_x + 2 * t_p) = \left(\frac{K}{R} + 2L\tau \right) \text{sec} = \boxed{\left(\frac{K + 2L\tau R}{R} \right) \text{sec}}$$



For Sliding Window ARQ:

$$\text{Optimal Window Size} = \left\lceil \frac{\text{Cycle Time (RTT)}}{\text{Transmission delay}} \right\rceil = \left\lceil \frac{K + 2L\tau R}{K} \right\rceil$$

Minimum number of bits required for sequence number field

$$= \left\lceil \log_2 [\text{Total number of sequences}] \right\rceil \text{ bits}$$

$$= \left\lceil \log_2 \left\lceil \frac{K + 2L\tau R}{K} \right\rceil \right\rceil \text{ bits}$$

Statements for Linked Answer Questions :

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

#Q. What is the minimum number of bits (l) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

[GATE 2009]

IIT-R, H.W.

- A** $l = 2$
- B** $l = 3$
- C** $l = 4$
- D** $l = 5$

#Q. Suppose that the sliding window protocol is used with the sender window size of 2^l where l is the number of bits identified in the previous question and acknowledgments are always piggybacked. After sending 2^l frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Identify the closest choice ignoring the frame processing time.)

- A** 16 ms
- B** 18 ms
- C** 20 ms
- D** 22 ms

[GATE 2009]

H.W.



Topic : Sliding Window ARQ



→ Types of Sliding Window Protocol :

1. Go Back N ARQ
2. Selective Repeat ARQ



Topic : Go Back N ARQ



→ Transmitter's transmitting window size = N

→ Receiver's receiving window size = 1

→ Total number of sequences = (N+1) [0 to N]

(N > 1)

Total number of sequences =

Transmitter's transmitting window size
+ Receiver's receiving window size

Sequence number $\leftarrow$ (Frame number) mod (N+1)



Topic : Go Back N ARQ

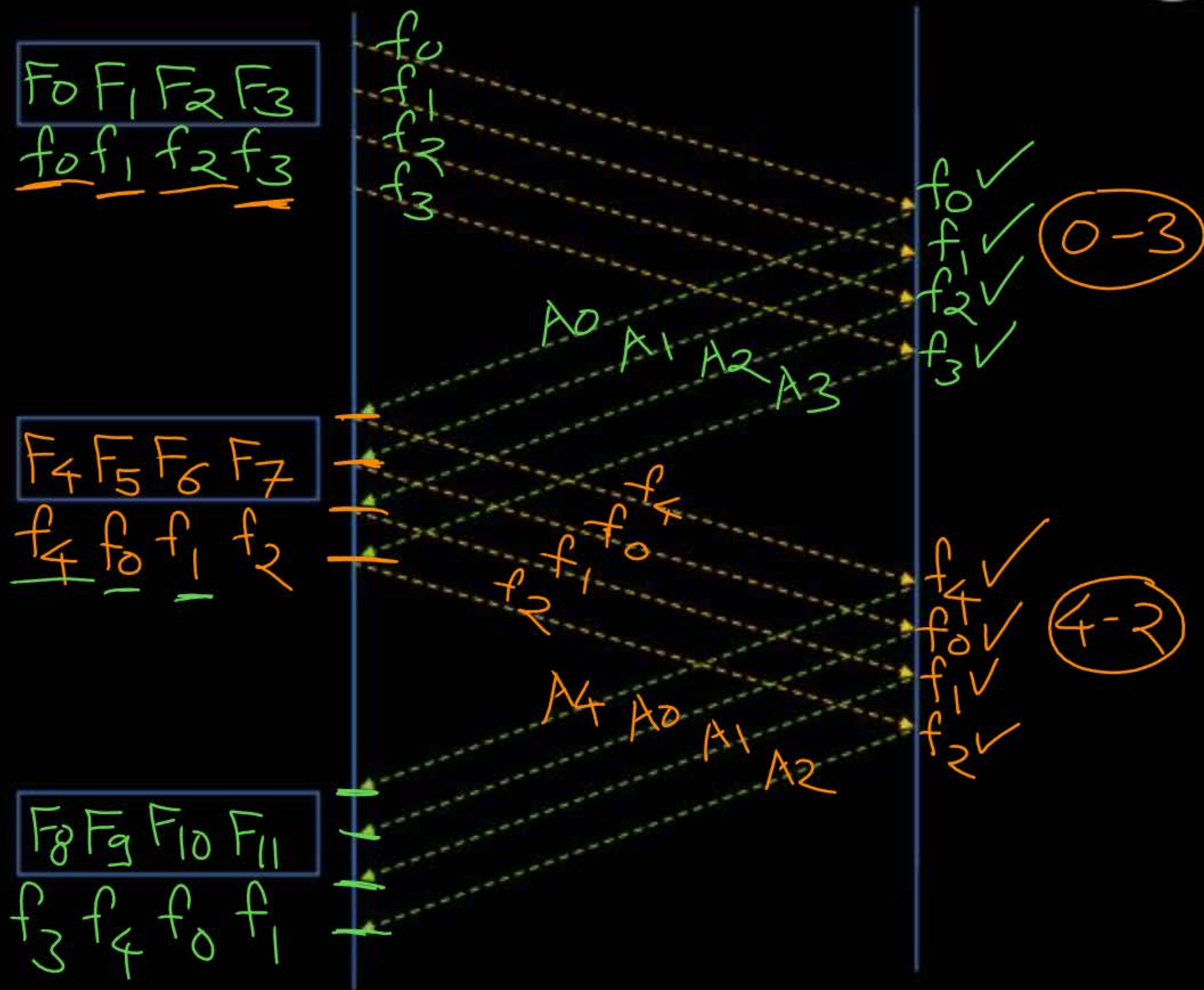


CASE I:

Go back 4 ARQ

$N=4$

Sequence Number = 0 to 4





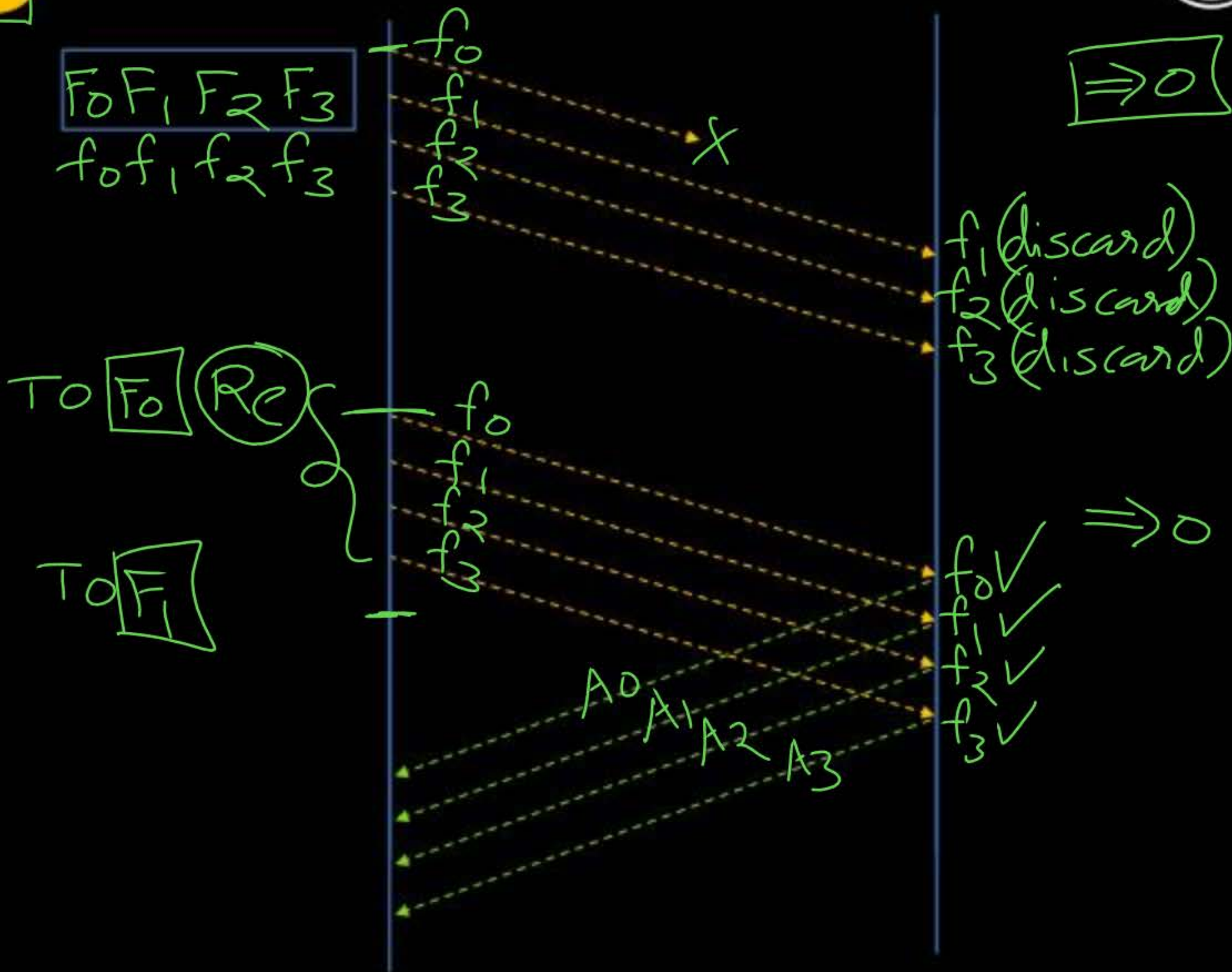
Topic : Go Back N ARQ



CASE II :

Go back 4 ARQ

Sequence Number = 0 to 4





Topic : Go Back N ARQ



CASE III:

Go back 4 ARQ

Sequence Number = 0 to 4

(Combine Ack)

$F_4 F_5 F_6 F_7$
 $f_4 f_0 f_1 f_2$

Transmitter

Receiver

$F_0 F_1 F_2 F_3$
 $f_0 f_1 f_2 f_3$

f_0
 f_1
 f_2
 f_3

X A0
X A1
X A2
A3

f_0 ✓
 f_1 ✓
 f_2 ✓
 f_3 ✓

f_4
 f_5
 f_6
 f_7



Topic : Go Back N ARQ



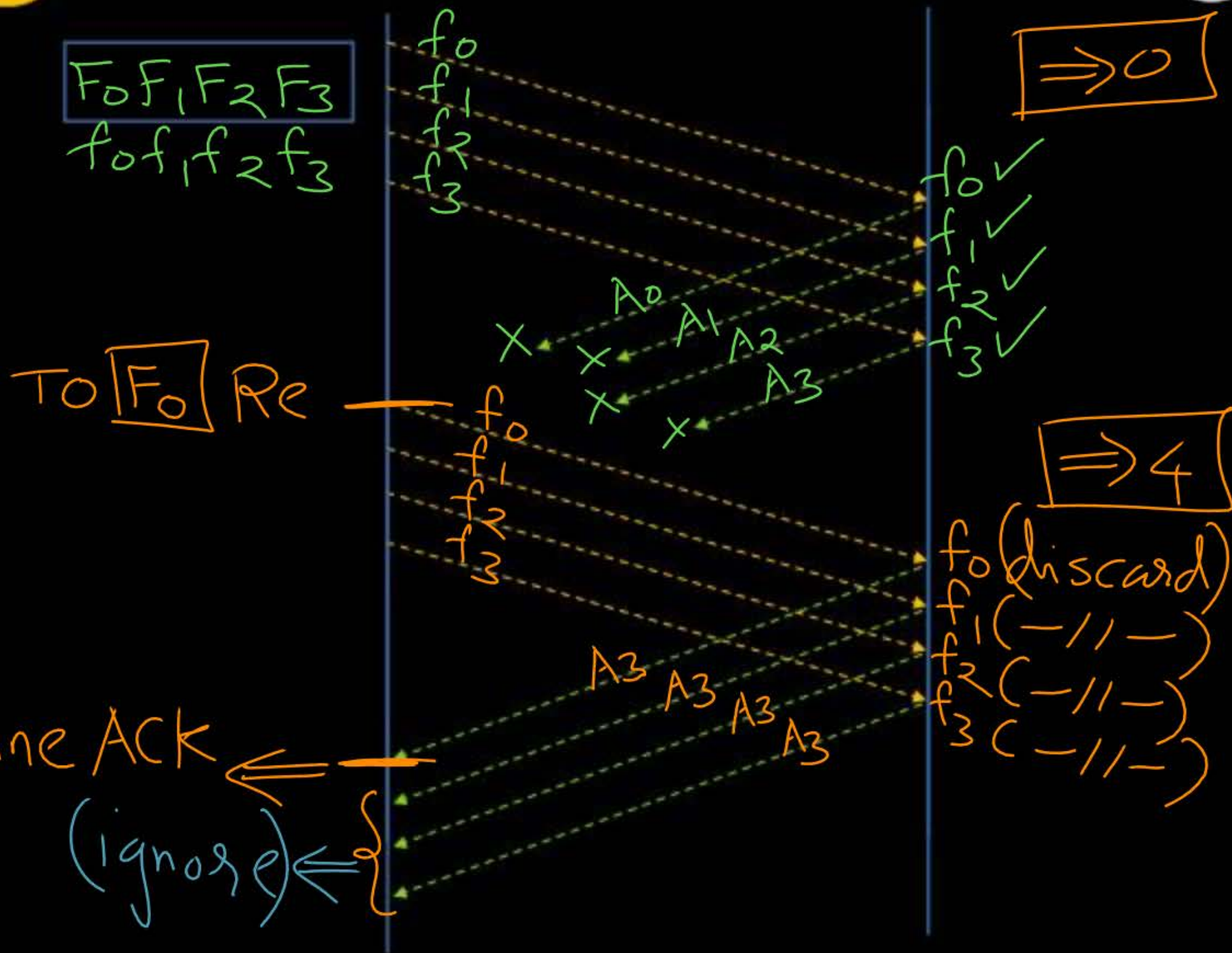
CASE IV :

Go back 4 ARQ

Sequence Number = 0 to 4

Receiver Always recognize (discard) duplicate frames.

Combine ACK
(ignore)





2 mins Summary



Topic

Sliding Window Protocol ✓

Topic

Go Back N ARQ



THANK - YOU

